🌿 MoodMate — Project Guide & Viva Cheat Sheet

For Developers Transitioning from Basic HTML & CSS to Modern React • Hackathon & Evaluation Ready

🌟 1. Project Overview in 30 Seconds

- **Project Name:** MoodMate — Your Everyday Happiness & Mental Reset Hub
- **Hackathon Theme:** "Building for Human Happiness & Connection"
- **What It Does:** A mental wellness and emotional companion web app where users can track daily feelings (8 distinct tactile moods), reflect with tailored micro-tasks and YouTube soundscapes, view weekly Joyful Bar Graphs, earn gamified "Growth Coins", practice 4-7-8 breathing, and chat with a real AI companion (**Lumi** 🌿, powered by Google Gemini).
- **Core Philosophy:** Non-punitive, non-clinical, and compassionate. Emotional ups and downs are treated like natural weather, not bugs to be fixed.

🛠️ 2. The Tech Stack (Explained for HTML/CSS Beginners)

If a judge asks: *"What technologies did you use to build this?"*, here is how you explain each tool simply:

Technology	What it is in simple terms	Why we used it over plain HTML/CSS
React 18	A JavaScript library for building user interfaces using reusable "components".	In HTML, to put a button on 5 pages, you copy-paste it 5 times. In React, you make `<Button />` once and reuse it. When data changes, React updates only that specific element without reloading the page.
Vite	An ultra-fast build tool and local development server.	Instead of opening a static file, Vite runs a local server (`http://localhost:5173`) and updates the browser in milliseconds whenever code changes.
Tailwind CSS	A modern utility-first CSS framework.	Instead of writing a huge `style.css` file with custom names, you use utility classes directly on HTML elements (e.g. `bg-emerald-500 text-white`

Technology	What it is in simple terms	Why we used it over plain HTML/CSS
		`rounded-xl p-4`). Fast, clean, and responsive.
React Router v6	Navigation manager for multi-page web apps.	In traditional websites, clicking a link loads a whole new HTML page. React Router switches the screen content instantly with zero reload (Single Page Application).
Google Gemini AI (3.8 Flash)	Google's official Large Language Model API.	Powers Lumi 🌿. Delivers genuine, empathetic mental wellness support and accurate coding assistance (C++, Python, algorithms) with multi-turn memory.
Framer Motion	Animation library for React.	Powers smooth page transitions, tactile button pops, and the soothing breathing circle pacer.
Canvas Confetti	Particle effects engine.	Celebrates task completions, streak milestones, and virtual hugs with bursts of colorful confetti.
LocalStorage	Browser-native permanent storage.	Keeps user accounts, mood logs, notes, and coins saved directly on the user's computer without requiring an expensive database server.

📂 3. Folder Structure (What lives where?)

```
Mood Mate/
├── index.html              # The single HTML entry point of the entire website
├── package.json            # Lists dependencies and npm scripts
├── vite.config.js          # Vite config & local API proxy for Gemini AI
├── vercel.json             # Deployment routing configuration for Vercel
├── .env                    # Stores secret keys (e.g. GEMINI_API_KEY)
│
├── api/                    # Cloud Serverless Functions
│   ├── chat.js             # Handles /api/chat in the cloud
│   └── health.js           # Server connectivity checks
│
├── server/
│   └── chatHandler.js      # Formats Gemini multi-turn messages & calls Google API
│
└── src/
    ├── main.jsx            # Mounts React into index.html (<div id="root">)
    ├── App.jsx             # Defines all page routes (/checkin, /analytics, /lumi, et
    │
    ├── context/            # Global Data Hubs
    │   ├── AuthContext.jsx # Handles user login, signup, avatar, and sessions
    │   ├── MoodContext.jsx # Stores 14-day mood history & check-in submissions
    │   └── WalletContext.jsx # Tracks Calm, Happy, and Growth Coins
    │
    ├── components/         # Reusable UI Blocks
    │   ├── Navbar.jsx      # Top header with live coin balances & streak
    │   ├── Footer.jsx      # Bottom links & demo data reset
    │   ├── LumiChat.jsx    # The full Gemini AI companion chat window
    │   └── MarkdownRenderer.jsx # Renders code snippets, copy button, & bold text
    │
    └── pages/              # The Full Screen Views
        ├── LandingPage.jsx   # Public landing page (/)
        ├── AuthGateway.jsx   # Login & Signup screen (/login)
        ├── CheckInPage.jsx   # 8-Mood tactile check-in & micro-tasks (/checkin)
        ├── AnalyticsPage.jsx # The Joyful Bar Graph (/analytics)
        ├── WalletPage.jsx    # Growth Coin rewards ledger (/wallet)
        ├── BreathePage.jsx   # 4-7-8 Zen Breathing Sanctuary (/breathe)
        └── CommunityPage.jsx # Kindness Wall & Virtual Hugs (/community)
```

🧠 4. How the Features Actually Work Behind the Scenes

A. How Does Data Stay Saved Without a Database?

MoodMate uses the browser's built-in `localStorage`:

- When you register, your profile is stored under `moodmate_users`.
- Your journal and check-ins are saved under `moodmate_data_<userId>`.
- Because data is partitioned by user ID, multiple users can use the same browser with isolated accounts!
- For judges, a **"Load Demo Profile"** button instantly loads 14 days of realistic logs.

B. How Does Lumi AI 🌿 Work?

1. User types a message in `LumiChat.jsx`.
2. The frontend sends the message, full conversation history, user name, and current mood to `/api/chat`.
3. `server/chatHandler.js` formats the conversation into Google's strict multi-turn payload (alternating `user` → `model`).
4. Google Gemini (`gemini-3.8-flash`) processes the prompt and returns the answer.
5. `MarkdownRenderer.jsx` formats code snippets with dark syntax styling and a working **"Copy Code"** button.
6. If the server encounters a network issue, Lumi automatically falls back to querying Gemini directly from the user's browser!

C. How Does the Joyful Bar Graph Work?

- Located in `src/pages/AnalyticsPage.jsx`.
- Reads the 7 days of the current week (Monday through Sunday).
- Assigns emotional energy scores from **1 to 8** (Happy = 7–8, Calm = 5–6, Stressed = 2–3).
- Renders rounded vertical columns dynamically colored by mood with interactive hover tooltips.

🎤 5. Common Judge & Viva Questions (Ready-Made Answers)

> **Q1: "Why did you build MoodMate instead of using a standard mood tracker like Daylio?"**
>
> **Answer:** *"Most trackers are clinical, punitive (inducing guilt for broken streaks), and treat sadness or stress like failures. MoodMate treats emotions like natural weather. It pairs tactile check-ins with somatic grounding (the 4-7-8 breathing pacer), micro-actions, and an intelligent AI companion (Lumi) that can help students de-escalate anxiety while also assisting them with college study and coding deadlines."*

> **Q2: "Is the AI companion real or are the responses hardcoded?"**
>
> **Answer:** *"Lumi is powered by a live integration with Google's official Gemini 3.8 Flash model. It has true multi-turn context retention—meaning you can ask it to write a C++ algorithm and then say 'explain how line 4 works', and it will remember. We also built custom Markdown rendering with one-click code copying and a dual-mode API bridge that works both server-side and browser-direct."*

> **Q3: "How does the app protect user privacy?"**
>
> **Answer:** *"MoodMate runs on a zero-leak local architecture. All personal reflections, mood logs, and credentials stay isolated inside the user's own browser storage (`localStorage`). No private journal entries are stored on third-party tracking databases."*

> **Q4: "What was the most challenging technical part of the project?"**
>
> **Answer:** *"Architecting the multi-turn Gemini conversation handler. Gemini's API strictly requires alternating turns starting with the user, so we wrote an intelligent normalizer in `server/chatHandler.js` that merges consecutive same-role messages, trims older history to prevent token overflows, and injects somatic empathy prompts while preserving technical reasoning capabilities."*

🏆 6. Presentation Checklist Before You Present

- ✅ Dev server running (`npm run dev`) or live link open.
- ✅ If using a fresh account, click **"Load Demo Profile"** so the Joyful Bar Graph has colorful data to show judges.
- ✅ Test prompt 1 ready in Lumi chat: Click the 💻 **C++ String Reverse** chip to demonstrate real AI speed.
- ✅ Test prompt 2 ready: Say *"Explain that line by line"* to prove multi-turn context retention.
- ✅ Click a mood on `/checkin` to showcase the particle explosion.
- ✅ Show the 4-7-8 Breathing Sanctuary (`/breathe`) for the human happiness theme connection.